

Received 18 December 2022, accepted 25 December 2022, date of publication 10 January 2023, date of current version 20 January 2023.

Digital Object Identifier 10.1109/ACCESS.2023.3235654

RESEARCH ARTICLE

Design and Control of a Size-Adjustable Pediatric Lower-Limb Exoskeleton Based on Weight Shift

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This work was supported by the Interreg 2 Seas Program 2014–2020 by the European Regional Development Fund (M.O.T.I.O.N Project) under Contract 2S05-038.

This work involved human subjects or animals in its research. The authors confirm that all human/animal subject research procedures and protocols are exempt from review board approval.

ABSTRACT Lower-limb exoskeletons have been proven to be beneficial for motor function disability patients, in both clinical rehabilitation settings and daily activities. However, exoskeletons for the pediatric field are still very limited. In this paper, a novel lower-limb exoskeleton design for children is presented. Based on the anthropometric data of the target group, the size of the exoskeleton is designed adjustable to suit children from 8 to 12 years old. It employs six active joints on the knee and hip, actuated by Brushless DC motors and Harmonic Drive gears. The controller is based on a finite state machine model and the weight shift between two feet to generate the gait trajectory. An innovative automatic step-triggering mechanism is also proposed, based on the feedback from the ground reaction force sensor on the foot. Two user interfaces on the exoskeleton and the host PC are designed for easily operating the exoskeleton by clinicians and engineers. Experiments have been conducted with three healthy volunteers following the predesigned rehabilitation session protocol. Results show that the exoskeleton can well follow the gait trajectory generated by the control algorithm. All volunteers can fulfill all tasks in the test protocol with reduced efforts and the steps are automatically triggered by the presented controller.

INDEX TERMS Exoskeletons, rehabilitation robotics, robot control, robot motion.

I. INTRODUCTION

Motor function disorders and impairments of children are known to be caused by neurological or neuromuscular diseases like cerebral palsy (CP), spinal muscular atrophy (SMA), and muscular dystrophy (MD) in addition to spinal cord injury (SCI). Whereas CP encompasses a group of permanent movement and posture development disorders, activity limitations are attributed to non-progressive disturbances of the developing fetal or infant brain [1]. Incidence of CP is from 2 to 2.5 per 1000 live births in Europe [2] and the resulting disabilities vary in dependence from partial to total. SMA is an autosomal recessive neurodegenerative

disease characterized by degeneration of the anterior horn cells of the spinal cord, atrophy of skeletal muscles, and generalized weakness [3]. SMA is the second most common fatal autosomal recessive disorder after cystic fibrosis, with an estimated incidence of 1 in 6,000 to 1 in 10,000 live births, and carrier frequency of 1/40–1/60 [4], [5]. MDs are chronic, causing progressive muscle weakness and decreasing activities. Nine main types of MD are recognized, among which Duchenne muscular dystrophy (DMD) is the most frequent in childhood with an incidence of 1/5000 boys [6]. SCI in the pediatric population is less frequent (approximately 3% to 5% of annual total SCI cases) but has significant psychological and physiological impacts [7].

Children with these prementioned disorders often show various motor function disabilities. The most common forms

The associate editor coordinating the review of this manuscript and approving it for publication was Yangmin Li¹.

among CP patients, for example, are spasticity, rigidity, and diminished coordination and motor control [1]. These disabilities often lead to a reduced ambulatory capability considerably limiting the life quality of patients. Moreover, as they grow, influence motor skill development will gradually become greater [9]. Pathological gaits consequently develop [8], [11]. The reasons for these pathological gaits are due to insufficient or excessive movement of the lower-limb joints that over the long-term lead to severe problems including joint pain or degenerative arthritis. Such conditions are due to elevated joint contact forces, bone deformities, lost balance from inadequate foot clearance, and increased energy consumption during walking [12]. Effective therapies to manage the deterioration of the musculoskeletal system are crucial for maintaining mobility while growing up. Conventional rehabilitation and medical treatments include physiotherapeutic interventions, functional neurostimulation, passive orthoses, botulinum toxin injections, and orthopedic surgery [10]. However, long-term use of passive orthoses can lead to greater weakness in the restricted muscle groups over time. Other non-invasive methods like physiotherapeutic interventions often demand intensive labor and their effectiveness highly depends on the experience of the physiotherapist and their physical strength.

Robotic assistive devices have already a proven track record in surgery, rehabilitation, and psychotherapy [13], [14], [15]. And in the rehabilitation field, wearable robotic systems or exoskeletons offer real advantages of physical strength, precision, and repeatability. Interest in this field is growing due to an increasing number of lower-limb disabled patients [16]. Many studies have been conducted on the design of exoskeletons for rehabilitation [17], [23], assisted load carriage [18], and enhanced walking efficiency [19]. Commercial devices have also had great success on the international market [20], [21], [22], although, most of these designs focus on adults, and are not adaptable for pediatrics due to their size and control logic.

Recently, the design of lower-limb exoskeletons for children have focused on the partial lower-limb structure or single degree of freedom (DOF). As an example, WAKE-up exoskeleton, an untethered powered knee–ankle–foot orthosis for children with CP characterized by position control aiding the knee and ankle during overground walking [24]. Furthermore, a powered exoskeleton for knee extension assistance as a treatment for crouch gait in children with CP has been designed [25]. Also, an ankle rehabilitation robot for traumatic brain injury children, with two modes of function, a passive mode forcefully stretching the ankle to the extreme dorsiflexion, and an active mode generating suitable torque during training [26], [27].

Nonetheless, exoskeletons with a full lower-limb structure and multiple actuated DOF for children are limited. Some commercially available exoskeletons have tried to modify adult products for the pediatric population [28], [29]. These versions are simply scaled down in size, whereas the control

algorithms remain unchanged. There are also other full lower-limb exoskeleton prototypes designed by some research institutes. Bayon et al. [30] designed an exoskeleton for CP children with 6 active DOFs at the hip, knee, and ankle, although requiring a smart walker to ensure balance. Similar designs with extra walker are also proposed for CP or SMA 2 children [31], [32]. However, the walker limited the mobility and the patient only follows a fixed gait pattern where the capability of maintaining balance is not trained. Other stand-alone exoskeletons for children with motor function disabilities are also proposed [33], [34], [35], but these designs are still primitive where only structure design and low-level control have been considered.

In this paper, MOTION, a hip-knee-ankle-foot exoskeleton for children with six active DOF including hip abduction/adduction, hip flexion/extension, and knee flexion/extension is presented. Size is readily adjusted for children with an integrated novel telescopic structure and clamping mechanism. IMU and ground reaction force sensors are added to the back and feet of the exoskeleton. In addition, a new gait trajectory control based on weight shift combining a novel automatic step-triggering mechanism is described and evaluated.

This paper is constructed as follows. The mechanical and electronic design of the exoskeleton is presented first. Followed by the kinematic analysis of the exoskeleton including both direct and inverse kinematic models. In the fourth chapter, the control method used on the exoskeleton is presented and then experimental results with three healthy candidates are shown in the next chapter. The final chapter concludes this paper with highlighted remarks and future perspectives.

II. MOTION LOWER LIMB EXOSKELETON

The aim of this research is to develop an exoskeleton that can augment the lower limb power of children, hence walking under a normal stable gait, as well as being able to sit and stand while wearing the device. To achieve these objectives, the design of the exoskeleton must be thoroughly considered both mechanically and electronically. Fig. 1 shows the prototype of the MOTION lower-limb exoskeleton in its sitting position.

A. STRUCTURAL DESIGN

Infants and youth go through drastic changes in body size, between 6 and 12 years of age. To design an exoskeleton for children, an analysis of anthropometric reference data was conducted [36], [37]. Precise and accurate estimations were obtained for minimum, average, and maximum lengths, and breadths of lower body segments of children between 8 and 12 years old. Thresholds exclusion of outliers were the 10th percentile as the minimum and 90th percentile as the maximum. Table 1 shows the minimum, average and maximum body dimensions of children between 8 and 12 years. Heights in this group of children vary from 1.16m to 1.73m.

The MOTION exoskeleton due to a structural design using carbon fiber tubes and telescopic cylinder mechanisms

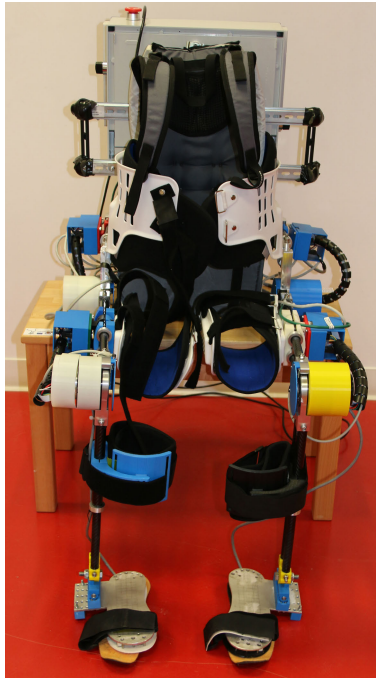


FIGURE 1. MOTION exoskeleton prototype in sitting position.

TABLE 1. Minimum, average, and maximum body dimensions of children at 8 and 12 years.

Age (yr)	8			12		
	Min	Average	Max	Min	Average	Max
H (mm)	1164	1316	1471	1374	1550	1731
ULL (mm)	278	318	356	338	387	436
TC (mm)	92	105	117	111	126	141
FL (mm)	175	199	225	206	235	265
LLL (mm)	263	300	340	319	362	408
SB (mm)	202	231	260	242	279	318
FB (mm)	63	72	82	73	83	95

H: Height, ULL: Upper leg length, TC: Thigh clearance, FL: Foot length
LLL: Lower leg length, SB: Seat breadth, FB: Foot breadth

is lightweight and adaptable to children of different sizes (Fig. 1). One end of the outer tube was vertically cut and with a clamp on its outside so that we can easily set it to the desired length. Additionally, quick-release skewers are used to make the adjustment process quicker and easier. These mechanisms on both waist and leg parts give maximum adaptability to different users. Furthermore, a joint-aligning mechanism assures hip joint compliance and natural hip internal/external rotation [38].

B. ACTUATION UNIT

For this research, we would like to design an exoskeleton that is able to totally provide power to assist children to walk. This means that in the fully powered mode, the user can walk without taking any effort. And as the rehabilitation progresses, assistance portion can be reduced.

In order to study the kinematic and dynamic characteristics of children's walking pattern, our partner in the MOTION project from the Research Group for Neurorehabilitation (eNRGy) at KU Leuven have conducted gait analyses on 31 typically developed children from eight to twelve years

TABLE 2. Gait data ranges for children from 8 to 12 years old.

	Sagittal Plane				Frontal Plane	
	Hip(F/E)		Knee(F/E)		Hip(A/A)	
	Mean	Max	Mean	Max	Mean	Max
Joint Torque (Nm)	22	126	14	65	18	63
Joint Power (W)	27	214	30	142	10	66
Joint Vel. (deg/s)	85	266	139	492	26	84

TABLE 3. Configuration of actuation unit in each joint.

	Rate Pow. (W)	Reduction ratio	Max. Tor. (Nm)	Max. Vel. (deg/s)
Hip(F/E)	220	100:1	150	300
Hip(A/A)	150	80:1	112	450
Knee(F/E)	150	50:1	70	720

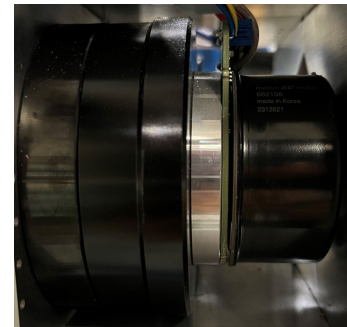


FIGURE 2. Maxon EC60 flat motor with harmonic drive[®] CPU-17A-M gear at hip A/A joint.

old. This dataset has the gait information with respect to the hip Flexion/Extension (F/E) and Abduction/Adduction (A/A), knee F/E and ankle Plantar/Dorsiflexion (D/P) joints. Table 2 shows the mean and maximum torque, power, and velocity at the joints which will be actuated by the exoskeleton.

The joint actuation unit is comprised of a 24V brushless motor (EC-flat, Maxon Motor AG, Switzerland) and strain wave gearing (CPU-17A-M, Harmonic Drive[®] AG, Germany). Maxon EPOS4 50/15 controllers are chosen as the driver of each motor, they support EtherCAT protocol and function as a slave in the network. To optimize weight torque and speed, different motors and reduction ratios were studied. Table 3 shows the configuration of actuation units in the three actuated joints per leg. It can be seen that all three actuation units can cover the maximum power, torque, and velocity obtained by gait analysis. Fig. 2 shows the actuation unit combination at the hip A/A joint.

C. SENSORY SYSTEM

Throughout MOTION, absolute encoders, pressure sensors, and IMUs are used.

For each actuated joint on the exoskeleton, a magnetic absolute rotary encoder is integrated (shown in Fig. 3(a)). The RLS AksIM-2 magnetic encoder is directly connected to the Maxon EPOS 4 motor driver under SSI communication

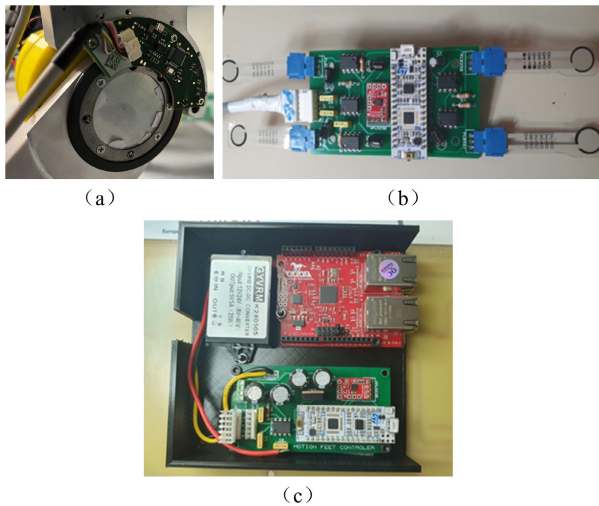


FIGURE 3. Sensors of the exoskeleton: (a) AksIM-2 magnetic absolute rotary encoder on the knee joint, (b) Foot sensor board (pressure sensor and IMU, and (c) Electronic board at the back (with IMU)).

protocol with a maximum resolution of 20 bits ensuring position sensing and position feedback control.

A sensor board (Fig. 3(b)) on each foot of the exoskeleton monitors ground reaction force and foot orientation. Four FlexiForce A201 force sensors measure pressure (front, back, left, and right). Additionally, an embedded IMU (MPU9250) acquires foot orientation. An STM32 Nano microcontroller manages signals and high-speed CAN bus data transmission.

The last sensor of the exoskeleton is the IMU on the back of the exoskeleton. This IMU is integrated into an electronic board into which an STM32 Nano MCU and an EtherCAT shield are also integrated as shown in Fig. 3(c). The electronic board can receive the sensory information of the two feet through CAN bus and the MCU will manage the receiving both the foot sensory information and the IMU on this board, then send all the sensory information to the main controller (EtherCAT master) via EtherCAT shield. The EtherCAT shield basically converts the electronic board into an EtherCAT slave.

D. CONTROL ARCHITECTURE

In order to control all six motors as well as collect sensor information in real-time, we set up the control system architecture of the exoskeleton as shown in Fig. 4. Generally, the system is separated into three parts: the actuation units, the sensory system, and the human-machine interface (HMI). And for the central control unit, we chose Speedgoat Unit real-time target machine as the onboard computer. The Unit target machine is the smallest model proposed by Speedgoat and it is designed for embedded deployment of rapid control prototyping. The Unit target machine can be served as the EtherCAT master and run the system with a frequency of 1kHz. One of the main advantages of Speedgoat is that the control Algorithm can be developed first in MATLAB Simulink[®] and then compiled in C executable file and deployed on the machine.

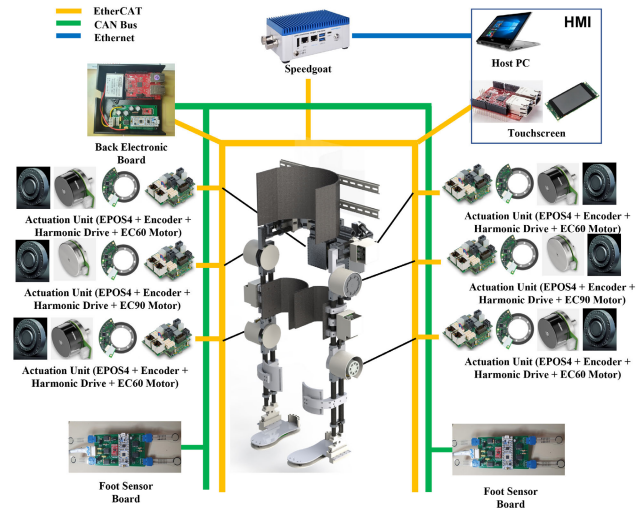


FIGURE 4. Control system architecture of the exoskeleton.

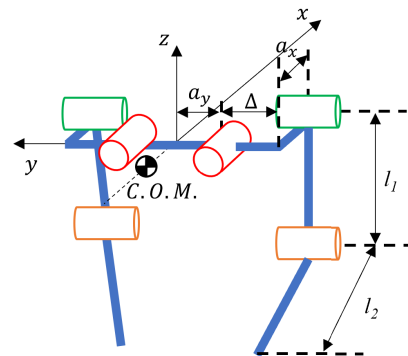


FIGURE 5. Kinematic chain of the exoskeleton.

For easily interacting with the exoskeleton by technical (engineer, developer) or nontechnical personnel (therapist, clinician), two types of HMI are provided in the control system as shown on the top left of Fig. 4. The first one is the Host PC connected to Speedgoat through ethernet. A relatively sophisticated application can be developed on this PC by using Matlab. Whereas the second one is a touchscreen-based user interface. It has been designed by using an STM32-based touchscreen development board and an EasyCAT Shield which converts this interface to an EtherCAT Slave. This touchscreen can work as a simple controller to send commands to the exoskeleton like the stand and sit, start walking, finish walking, etc.

III. EXOSKELETON KINEMATICS

In order to control the exoskeleton in joint space, we have to construct the kinematic model of the exoskeleton. The exoskeleton presented in this paper has 6 actuated DOFs including lateral movement and similar to the biped robots, exoskeletons have switching contact points (change of support leg), all these factors will make the kinematics of the exoskeleton quite complicated. The kinematic chain of the exoskeleton is presented in Figure 5. In this section, we are going to step by step build both the direct and inverse kinematic model of the exoskeleton.

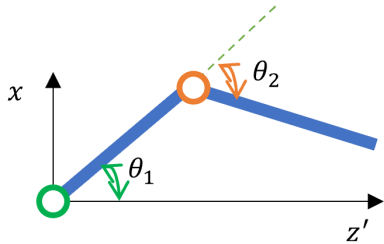


FIGURE 6. Kinematic chain of a single exoskeleton's leg in the plane of thigh and shank.

A. DIRECT LEG KINEMATICS

We can at first choose a single leg to analyze the direct kinematics, and in this case, the leg can be seen as a 3-DOF manipulator. We chose the middle of the back as the global reference origin and the kinematic chain of a single exoskeleton's leg in the plane of thigh and shank can be seen in Figure 6.

In the plane formed by the thigh and the shank, the x-axis is the same as the global reference, but since the leg has also abduction/adduction movement, here we use z' to present the axis perpendicular to the x-axis in this plane. And the position of the foot can be presented in this plane as follows:

$$\begin{cases} z'_f = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2) \\ x_f = a_x + l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2) \end{cases} \quad (1)$$

Now let us take the movement in the lateral direction into consideration. Figure 7 shows the kinematic chain of the right leg of the exoskeleton in the frontal plane. According to the geometric relationship, the right foot position in the frontal plane can be calculated by the following equation:

$$\begin{cases} y_f = -a_y - z'_f \sin \theta_0 - \Delta \cos \theta_0 + \frac{1}{2} w_f \\ z_f = -z'_f \cos \theta_0 + \Delta \sin \theta_0 \end{cases} \quad (2)$$

And symmetrically, the coordinates of the left foot in the frontal plane are presented by:

$$\begin{cases} y_f = a_y + z'_f \sin \theta_0 + \Delta \cos \theta_0 - \frac{1}{2} w_f \\ z_f = -z'_f \cos \theta_0 - \Delta \sin \theta_0 \end{cases} \quad (3)$$

With all the pre-mentioned equations 1 to 3, we have constructed the direct kinematics of the exoskeleton.

B. INVERSE LEG KINEMATICS

Through eliminating z'_f and and by using the tangent half-angle formula $\sin \theta_0 = \frac{2 \tan(\theta_0/2)}{1 + \tan^2(\theta_0/2)}$, $\cos \theta_0 = \frac{1 - \tan^2(\theta_0/2)}{1 + \tan^2(\theta_0/2)}$, we can have the following polynomial function:

$$\begin{cases} (y_f - a_y)p^4 + 2z_f p^3 - 4\Delta p^2 + 2z_f p - (y_f - a_y) = 0 & \text{(left leg)} \\ (y_f - a_y)p^4 + 2z_f p^3 + 4\Delta p^2 + 2z_f p - (y_f - a_y) = 0 & \text{(right leg)} \end{cases} \quad (4)$$

where $p = \tan(\theta_0/2)$.

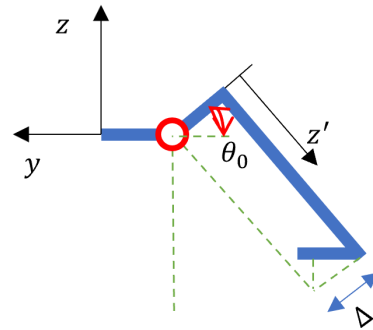


FIGURE 7. Kinematic chain of the right leg of the exoskeleton in frontal plane.

Hence by solving these equations, we can easily obtain the hip abduction/adduction angle by:

$$\theta_0 = 2 \arctan(p) \quad (5)$$

By substituting θ_0 into (2) and (3), we can have

$$\begin{cases} z'_f = \frac{-z_f - \Delta \sin \theta_0}{\cos \theta_0} & \text{(left leg)} \\ z'_f = \frac{-z_f + \Delta \sin \theta_0}{\cos \theta_0} & \text{(right leg)} \end{cases} \quad (6)$$

Then the hip and knee flexion/extension angles can be solved by:

$$\begin{cases} \theta_2 = -\arccos\left(\frac{z_f'^2 + x_f^2 - l_1^2 - l_2^2}{2l_1 l_2}\right) \\ \theta_1 = \arctan\left(\frac{x_f}{z'_f}\right) - \arctan\left(\frac{l_2 \sin \theta_2}{l_1 + l_2 \cos \theta_2}\right) \end{cases} \quad (7)$$

With equations 4 to 7, we have obtained the inverse kinematic model of the exoskeleton.

IV. CONTROL STRATEGY

The control strategy of lower-limb exoskeletons (low-level) can be generally separated into two categories: position control and torque control. Although torque control has advantages like supporting the transparent mode and being able to partially provide assistance force, it needs a rather complicated joint-level design (i.e. strain gauge or serial elastic actuator) for the estimation of the joint's output torque. In this paper, our control strategy is based on position control at the joint level.

A. FINITE STATE MACHINE

For patients using an exoskeleton to do rehabilitation training, they should first learn how to use the exoskeleton, since most of the existing exoskeletons should be used with crutches and patients need to know how to manipulate their center of mass with the help of crutches. This learning process may take quite a long time and it also needs the prerequisite that the patient has strong enough upper-body strength to hold a large part of their body weight. In order to let the patient easily shift their center of mass (COM) between two feet, a finite

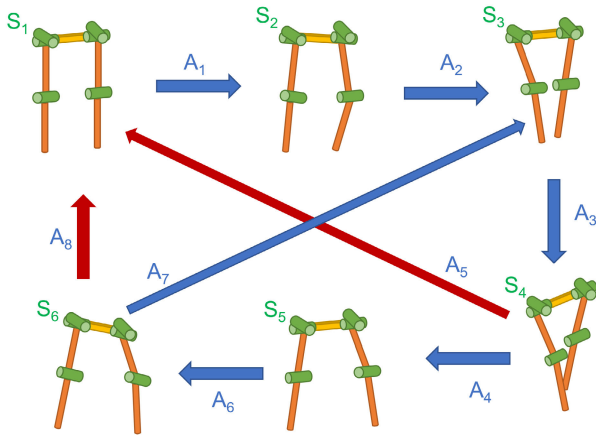


FIGURE 8. Finite state machine model of the exoskeleton.

state machine (FSM) model is proposed in which the COM shifts during the double-stand phase.

The general scheme of the proposed FSM model can be seen in Figure 8, where the exoskeleton's gait is separated into 6 different states. These states are defined as follows:

- *State 1 (S_1)*: The exoskeleton is in stand-up position and the feet are parallel. Usually, this is an initial or final state (after the user stands up from the chair or before the user sits down on the chair).
- *State 2 (S_2)*: Both feet are still parallel, however, the joint configuration of the exoskeleton is changed to let the COM approach the left foot.
- *State 3 (S_3)*: The feet are no longer parallel where the right foot is at the front and the left foot is at the rear. The COM of the exoskeleton is located in the center of the support polygon formed by two feet.
- *State 4 (S_4)*: The right foot is still at the front and the left foot is still at the rear, but the COM is moved from the center of the support polygon to the right foot.
- *State 5 (S_5)*: Symmetric to S_3 , the feet of the exoskeleton changed to left foot at the front and left foot at the rear. And the COM is still located in the center of the support polygon.
- *State 6 (S_6)*: Symmetric to S_4 , the feet of the exoskeleton changed to the left foot in the front and left foot in the rear. And instead of on the right foot, in this state, the COM will be located on the left foot.

Transitions between different states need to go through some well-designed actions. Details of these actions are as follows:

- *Action 1 (A_1)*: The exoskeleton is in the double stand position and two feet parallel, this action will gradually shift the COM (both in the frontal and lateral direction) to the left foot while holding both feet on the ground.
- *Action 2 (A_2)*: The right leg starts to swing and move the right foot forward while the left leg is standing on the ground as the supporting pivot. For doing this action, the right leg should support as less as possible of weight to facilitate the movement.

- *Action 3 (A_3)*: COM will be gradually shifted to the right foot while the left foot and right foot are both on the ground.
- *Action 4 (A_4)*: The left leg starts to swing and move the left foot forward to make an entire step. The right leg stays on the ground as the supporting pivot. Similar to A_2 , the left leg needs to be as much as possible liberated in order to easily move.
- *Action 5 (A_5)*: This action is an alternative action when exoskeleton is in S_4 . Instead of moving the right foot for an entire step, this action will let the exoskeleton walk a half step, and the two feet will end up in a parallel position (back to S_1).
- *Action 6 (A_6)*: COM will be gradually shifted to the left foot while the left foot and right foot are both on the ground.
- *Action 7 (A_7)*: The right leg starts to swing and move the right foot forward to make an entire step. The left foot stays on the ground as the supporting pivot. This action brings the exoskeleton back to a previous state (S_3) so that a closed loop for continuous walking can be formed: $S_6 \Rightarrow S_3 \Rightarrow S_4 \Rightarrow S_5 \Rightarrow S_6$.
- *Action 8 (A_8)*: This action is an alternative action when exoskeleton is in S_6 . The right leg walks only a half step to let the exoskeleton return to its initial state (S_1).

B. GAIT PLANNING CONSIDERING COM SHIFT

Previously we have explained how the exoskeleton's states can be changed by applying corresponding actions. And these actions are basically actuated by executing different gait trajectories for both legs. However, the gait should be carefully planned since the objective of each action is different as well as the support state (double-foot support or single-foot support). For example, in A_3 , the two feet need to stay on the ground while the center of mass is moving toward the right foot, and in A_4 , only the right foot stays on the ground whereas the left foot leaves the ground and moves forward.

During the rehabilitation training with an exoskeleton, one of the most challenging parts for the patient is the shifting of their COM to a suitable position so that they can easily lift up the swing leg and move forward. And this could be done by changing the joint's configuration while the user is in the double stance phase. We can first plan the trajectory of the COM and then calculate the joint configuration by using the inverse kinematic model that we conducted in the previous section.

The COM shift is generally through manipulating its position projected on the transverse plane. In A_1 , we need to shift the COM on the left foot and hence the COM position on the x-y plane is:

$$\begin{cases} p_{COM,1}^x = 0 \\ p_{COM,1}^y(t) = \delta(t, t_f, 0, y_{COM}^{trans}) \end{cases} \quad (8)$$

where $\delta(t, t_f, 0, y_{COM}^{trans})$ is the minimum jerk trajectory function with zero initial and final velocity and acceleration and

y_{COM}^{trans} is the final COM shift offset in y direction. The minimum jerk trajectory function is a commonly used method in the robotic field for generating both smooth and energy-efficient movement, it is formulated by a fifth-order polynomial equation [39].

After the COM has been well shifted on the support leg, we can start to move our body forward as well as the COM position. In order to guarantee maximum stability, we would like to keep the COM in the middle of the support polygon (i.e. the middle point of two feet). Hence in A_2 , the COM trajectory in the x-y plane is:

$$\begin{cases} p_{COM,2}^x(t) = \delta(t, t_f, 0, \frac{l_s}{2}) \\ p_{COM,2}^y(t) = \delta(t, t_f, y_{COM}^{trans}, 0) \end{cases} \quad (9)$$

where l_s is the predefined step length, it is usually a case-by-case setting and should be defined by the physiotherapist.

Then, for the next step, the COM position needs to be shifted to the right foot. And in A_3 , there is a difference from A_1 in terms of COM shift that the COM needs to be moved in both later and longitudinal directions. So the COM trajectory in A_3 is defined as:

$$\begin{cases} p_{COM,3}^x(t) = \delta(t, t_f, \frac{l_s}{2}, \frac{l_s}{2} + x_{COM}^{trans}) \\ p_{COM,3}^y(t) = \delta(t, t_f, 0, -y_{COM}^{trans}) \end{cases} \quad (10)$$

where x_{COM}^{trans} is the final COM shift offset in x direction.

After A_3 brings to S_4 , we can choose two options. The first one is to take A_4 to make a full step and continue the training session and the second is to take A_5 and make a half step and end the training session. If we take A_4 , the COM trajectory will be:

$$\begin{cases} p_{COM,4}^x(t) = \delta(t, t_f, \frac{l_s}{2} + x_{COM}^{trans}, \frac{3l_s}{2}) \\ p_{COM,4}^y(t) = \delta(t, t_f, -y_{COM}^{trans}, 0) \end{cases} \quad (11)$$

, and if take A_5 , the corresponding COM trajectory is:

$$\begin{cases} p_{COM,5}^x(t) = \delta(t, t_f, \frac{l_s}{2} + x_{COM}^{trans}, l_s) \\ p_{COM,5}^y(t) = \delta(t, t_f, -y_{COM}^{trans}, 0) \end{cases} \quad (12)$$

The rest of the actions are symmetrical to the aforementioned action. For example, A_6 is symmetrical to A_3 , A_7 is symmetrical to A_4 , and A_8 is symmetrical to A_5 . And for these pairs of symmetrical actions, the joint angle trajectory of one leg of one action is the same as the trajectory of the opposite leg of the other action.

The step height of the gait is defined as a semi-eclipse trajectory where the step height reaches the maximum in the middle of the step.

C. AUTOMATIC STEP TRIGGERING

For making the system fully automatic, a step-triggering strategy has been designed. In a normal human walking gait, a step is more or less triggered by ankle push-off which will

accelerate the forward speed of COM and also displace the COM more forwardly. However, the exoskeleton presented in this paper does not have actuated ankles and hence the push-off force from the ankle joint cannot be powered by the exoskeleton. In the previous section, we introduced a gait control strategy that includes a COM shift phase to move the COM on the standing leg. But in the actual use, because of the undesired relative movement between the user and the machine, the actual COM position may not be well shifted on the standing leg. And in this scenario, if we start the swing motion, the swing leg will not be able to lift up and also the user may lose balance and fall down.

In order to make sure the COM has been well shifted to the standing leg as well as to trigger each step automatically, an automatic step-triggering strategy based on the ground reaction force on the swing leg is proposed. The main idea of this strategy is to monitor the change of ground reaction force of the swing leg during the COM shift actions (A_1 , A_3 , and A_6), if the force is sufficiently decreased compared to the start of the action, we can assume that the COM transition is done and the step can be triggered. And this decrease is defined as follows:

$$\gamma_{GRF,i}^t = \frac{F_{GRF,i}^0 - F_{GRF,i}^t}{F_{GRF,i}^0} \quad (i = 1, 3, \text{ or } 6) \quad (13)$$

where $F_{GRF,i}^0$ is the ground reaction force of the swing leg at the start of the action and $F_{GRF,i}^t$ is the value after the action is finished.

The step is triggered just by a simple judgment. After the COM shift action is finished, the controller will keep monitoring the $\gamma_{GRF,i}^t$ and if it is larger than a preset threshold, the step will be triggered.

D. IMPLEMENTATION USING SIMULINK REAL-TIME

All the control algorithms presented previously are programmed via Simulink Real-Time of MATLAB®. Simulink Real-Time is a rapid control prototyping in which you can easily model the control system by using varieties of blocks from Simulink and then compile it to real-time applications that can run on Speedgoat real-time target computers.

The Simulink model of the exoskeleton's control system is shown in Fig. 9. This model consists of three main parts, EtherCAT network initialization and error management, SDO communication service to parameter settings like the PID gains, and PDO communication which contains the core of the control algorithm and runs in real-time with a frequency of 1kHz.

V. USER INTERFACES OF THE EXOSKELETON

In order to let physicians or engineers easily control the prototype, two user interfaces on the host computer and the onboard touchscreen have been designed respectively. The reason to have two interfaces is to meet the needs of different staff. Application engineers have a relatively sophisticated user interface that can deal with the connection, user information, gait generation, surveillance of motor parameters,

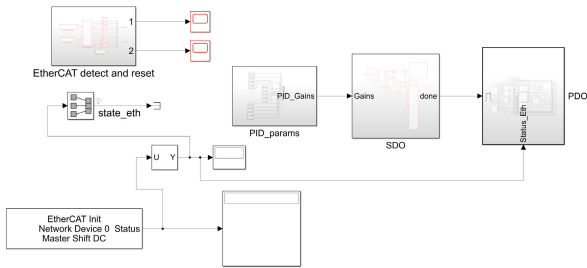


FIGURE 9. Simulink model of the control system.

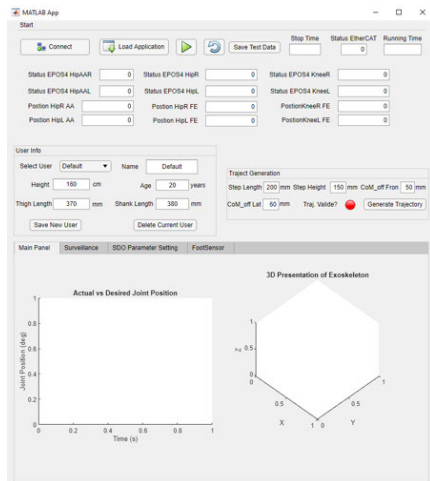


FIGURE 10. Matlab application user interface on the host machine.

etc. However, physicians need a much simpler interface with fewer parameters to adjust.

A. MATLAB APPLICATION

The user interface application on the host machine is designed by using MATLAB[®] App Designer, shown in Fig. 10. This application has several useful modules for the engineer to monitor the status of the exoskeleton as well as easily adjust parameters during the test. On the top part of the application, we have several buttons to manage the connection between the host PC and the Speedgoat real-time target machine. And followed by that is the streaming EPOS4 drivers status and joint angles, the engineer can monitor these parameters to see if the driver is in an error state. This application also has a user management system in which you can use saved users' ergonomic information or create a new user in the database. Then based on the loaded information, the embedded gait generation module can generate a suitable gait base on the user's lower limb length as well as the predefined gait length and height. This application also has some other functionalities like a digital mock-up of the exoskeleton, PID gains adjustment, and some steaming plots like the desired vs. actual joint positions. After the test is finished, you can also save all the test data into a ".csv" worksheet file.

B. TOUCH SCREEN USER INTERFACE

Apart from the application on the host machine, another user interface is designed based on an embedded touch screen

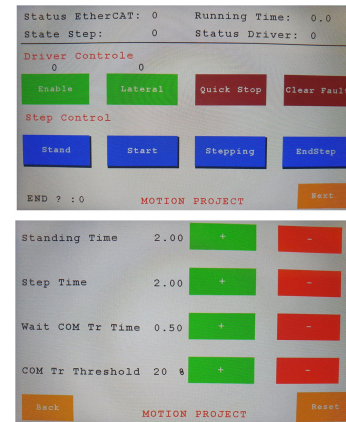


FIGURE 11. Touchscreen user interface on the back of the exoskeleton.

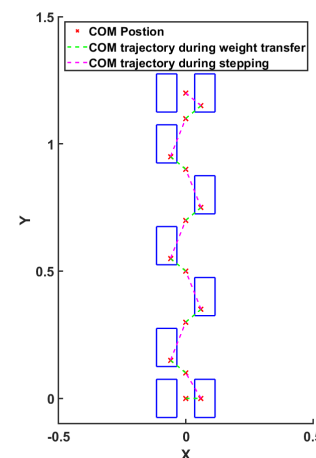


FIGURE 12. The prescribed gait trajectory in the horizontal plane.

which is fixed on the back electronic box of the exoskeleton. Fig. 11 shows the two-page user interface on the touchscreen. It can be seen that this interface is much simpler compared to the host PC application, the interface contains only a few buttons to control the exoskeleton like enabling the motors, starting the walk, and some parameter adjustments like the step time and step triggering threshold.

VI. EXPERIMENT AND RESULTS

Based on the previously presented exoskeleton design and control method, an exoskeleton prototype has been fabricated. And for evaluating the feasibility of the proposed design, experiments have been conducted with three healthy adults at the initial phase.

A. HEALTHY SUBJECT TESTS

For validating the functionality of the proposed exoskeleton design in this paper, tests have been conducted with volunteers. And since the proposed exoskeleton is only a preliminary design and the tests only intend to validate some of the basic functionalities, instead of testing it directly on child patients, we have found some healthy subjects with sizes inferior to the maximum allowable size defined in the

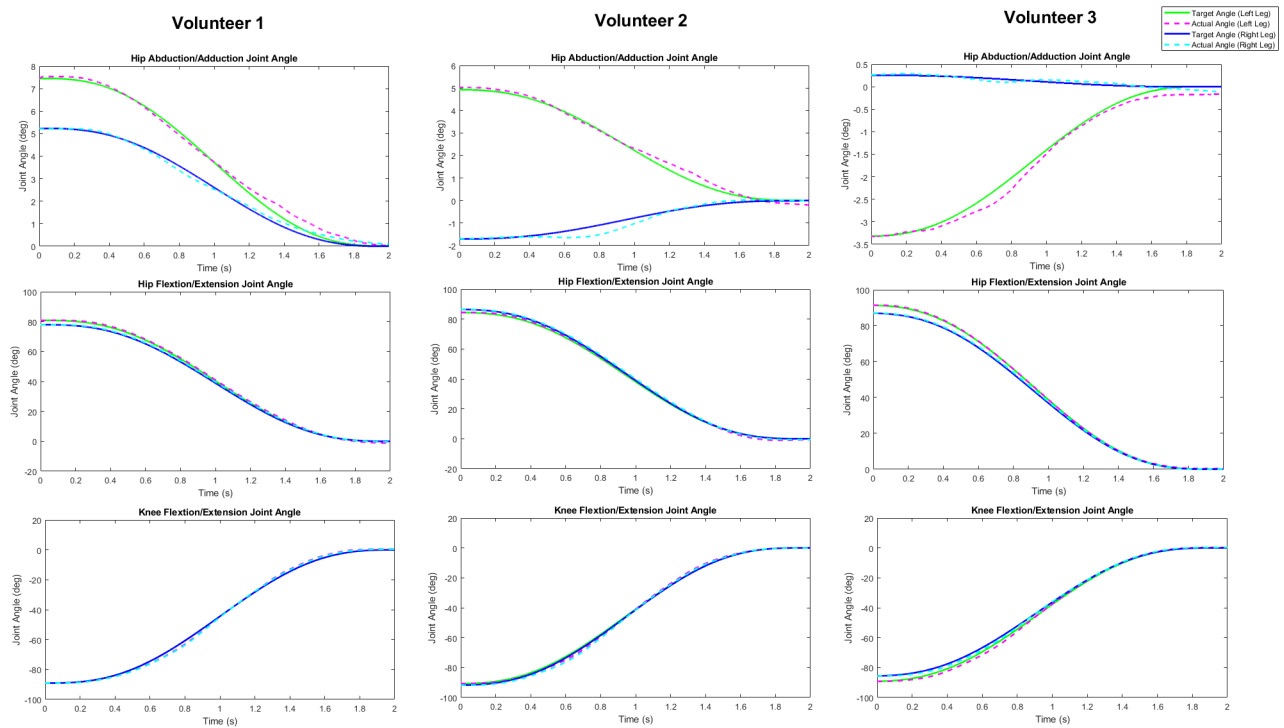


FIGURE 13. Joint trajectory tracking during standing phase.

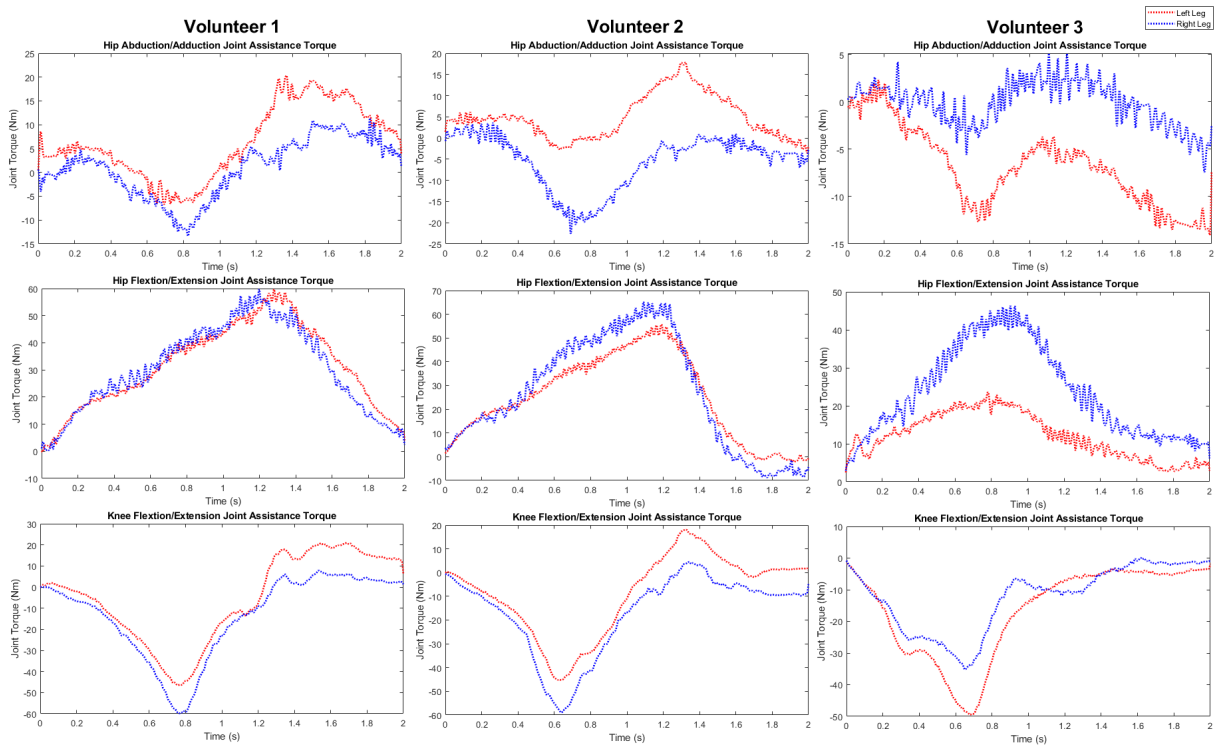


FIGURE 14. Joint assistance torque during standing phase.

design criteria. Table 4 shows the anthropometric data of three volunteers involved in the tests.

Before starting the tests, we informed the participants of the functionality of the exoskeleton, the test protocol, and potential risks. Signed informed consents were obtained.

During the tests, all the participants are required to follow a test protocol representing a classical rehabilitation training session. This protocol is defined as follows:

- 1) Put the exoskeleton on a bench in the sit position and open all the scratch tapes.



FIGURE 15. Snapshots of the volunteer walking during the experiment.

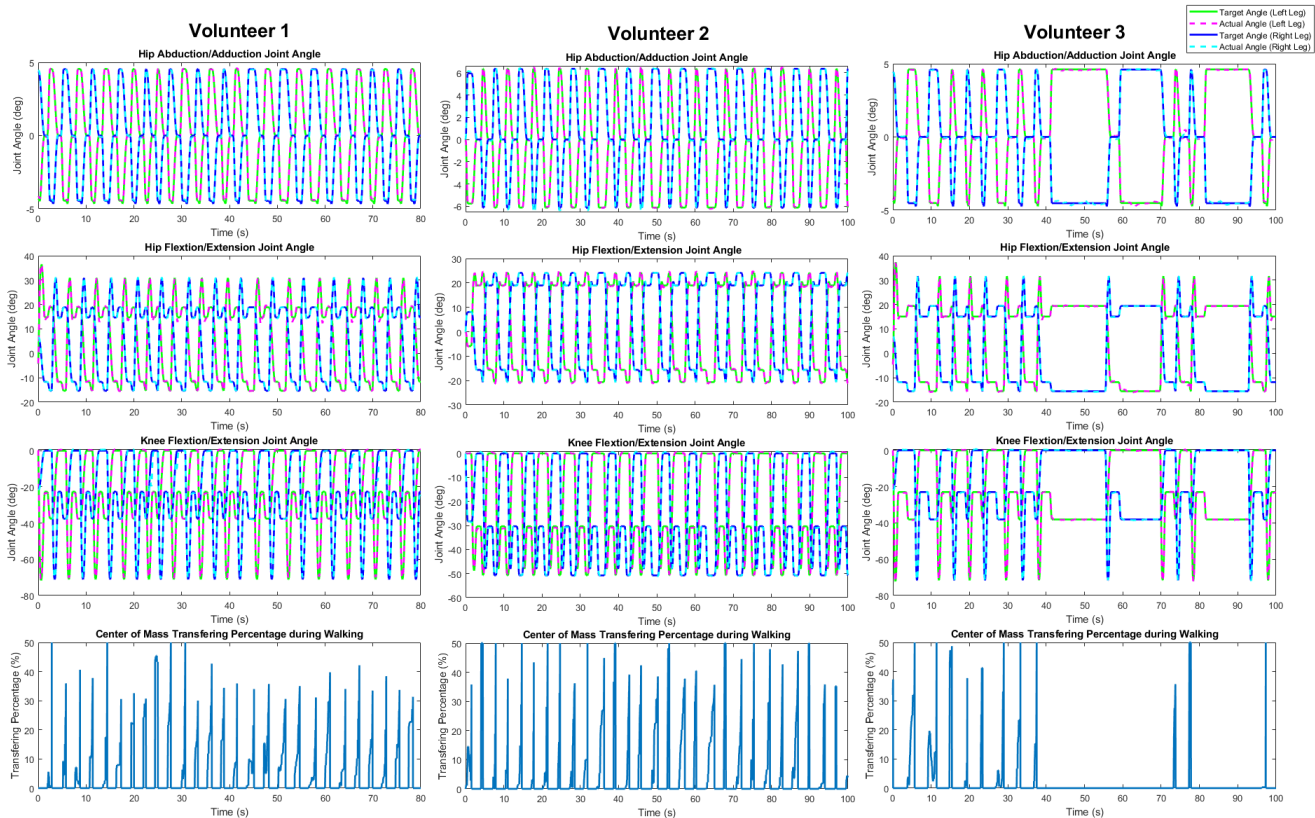


FIGURE 16. Joint trajectory tracking and COM shift ratio during walking phase.

- 2) Participant sits into the exoskeleton, then adjust the size of the exoskeleton to align all the joints and don the exoskeleton by closing the scratch tapes.
- 3) Power up the exoskeleton and wait until the exoskeleton is in the ready state.
- 4) Send stand command to the exoskeleton, participant stands up with the help of the exoskeleton.
- 5) Send start walking command to the exoskeleton, participant walks 10 meters with the exoskeleton.
- 6) After 10-meters walking, send the end walking command to the exoskeleton, participant stop walking with two feet in parallel position.
- 7) Send sit command to the exoskeleton and the participant sit down on the bench with the help of the staff.
- 8) Participant doffs the exoskeleton on the bench and leaves the exoskeleton. The test session is finished.

TABLE 4. Anthropometric data of the volunteers.

Volunteer	Height (cm)	Weight (kg)	Thigh Length (cm)	Shank Length (cm)
1	156	45	40	42
2	157	52	40	41
3	151	50	40	40

In the walking phase of the tests, the exoskeleton is encoded with the same gait trajectory. This gait trajectory is generated based on the method proposed in section IV with the following parameters:

- Step length l_s : 0.2 meter
- Step height h_s : 0.2 meter
- COM shift offset in x direction x_{COM}^{trans} : 0.05 meter
- COM shift offset in y direction y_{COM}^{trans} : 0.06 meter
- Step time T_s : 2 seconds

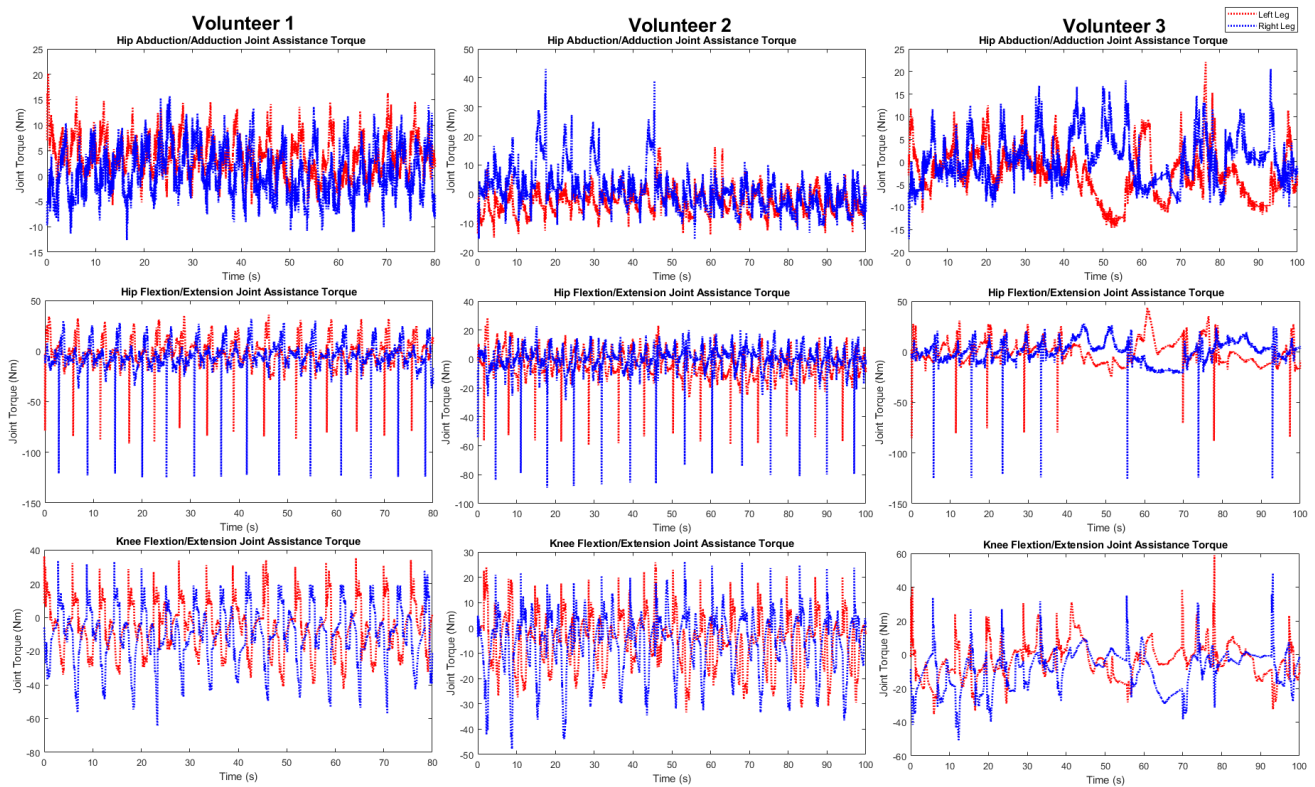


FIGURE 17. Joint assistance torque during walking phase.

Fig. 12 shows several steps of the gait trajectory used in the tests in which both footstep positions and COM trajectory are shown. It can be seen that before each step, the COM is deliberately moved on the standing foot to guarantee a more smooth state transit (i.e. stepping).

B. TEST RESULTS

1) STANDING PHASE

Fig. 13 shows the test results of the joint trajectory tracking for all the actuated joints during the standing phase. It can be seen that the control system of the exoskeleton generates smooth reference trajectories to bring the exoskeleton to the standing configuration (all the joints' angles are equal to 0 deg) and the exoskeleton tracks well the trajectory with only little tracking error. The torques produced by the actuators during the standing phase are shown in Fig. 14. Since the joints on the sagittal plane are the main power source during the stand-up task, the joint torques of these joints are much higher than the hip A/A joints.

2) WALKING PHASE

Fig. 15 shows the snapshots of one participant walking with the exoskeleton. With the COM shift strategy, the tester can easily shift their weight on the supporting leg and consequently be easier to move the swing leg forward. Fig. 16 shows the joint trajectory tracking and COM shift ratio during the walking phase. It can be seen that the joint reference trajectories are different for different users since these

trajectories depend on the lower-limb size of the user and steps are automatically triggered after the COM shift ratio surpasses the predefined value. Exoskeleton tracks well the reference trajectory for all three participants, and for volunteer 1 and volunteer 2, they manipulate quite well their COM while volunteer 3 has some difficulty changing the COM between two feet. This is normal for some users with no experience in manipulating their COM deliberately while doing sports or other activities. The torques produced by joint actuators during the walking phase are shown in Fig. 17. The hip flexion/extension joint is a major source for walking as it pushes the user to move forward. And all the produced torques are within the value we have estimated in Table 2.

VII. CONCLUSION

This paper presents the work, observations, and results concerning the design and performance of the MOTION exoskeleton, a novel size-adjustable pediatric lower-limb exoskeleton to fit the size of children from 8 to 12 years old. Brushless DC motors and Harmonic Drive gears are used as actuation units ensuring an adequate power supply. The industrial field bus system EtherCAT is used as the communication protocol of the proposed exoskeleton that allows a high communication frequency (1kHz for the proposed design). The control of the MOTION exoskeleton is based on an FSM-based control algorithm that considers the COM shift, allowing a smooth COM shift during the double stance phase. An automatic step-triggering algorithm based on the COM

shift ratio is also included in the control system allowing safe stepping. Additionally, for simplifying the operation of the MOTION exoskeleton, two user interfaces are designed for engineers and therapists respectively, considering their different needs. Preliminary tests have been conducted in a laboratory environment with three healthy adults of sizes similar to children's. Test results showed that all the participants can fulfill the predefined test protocol while wearing the exoskeleton, and the exoskeleton tracked well the reference trajectory during both the standing and walking phases. These results validated the feasibility of the MOTION lower-limb exoskeleton for being a rehabilitation device for the pediatric population.

In the future, more tests will be conducted on healthy children and children with neurological disorders in the clinical environment. These tests will help to fine-tune the exoskeleton for children and further on validating the functionality. Additionally, a feedback control algorithm ensuring stability will be designed, and visual sensors like LiDAR could be added in order to make the exoskeleton more autonomous while being used in a more challenging environment (ex., climbing stairs).

ACKNOWLEDGMENT

The authors would like to express their gratitude to all the participants who evolved in the tests.

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